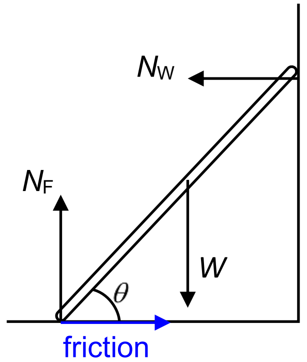
8 **B** sketch in order visualize



$$N_F = W$$
$$N_W = \text{friction}$$

pivot at wall

$$W\left(\frac{\cancel{L}}{2}\cos\theta\right) + (\text{friction})(\cancel{L}\sin\theta)$$
$$= N_F(\cancel{L}\cos\theta)$$

$$N_F = \frac{W}{2} + N_W\tan\theta$$
$$N_W\tan\theta = N_F - \frac{W}{2}$$

9 **D** tension in string 1 provides centripetal force for 3 masses so must be largest magnitude: